Hierarchical Ordinal Framework for Automated Movie Censorship Using Full-Length Scripts

Pratik N. Kalamkar

Department of Computer Science and Engineering
Shri Jagdishprasad Jhabarmal Tibrewala University
Jhunjhunu, Rajasthan, India
pratik2kcn@gmail.com

Yogesh Kumar Sharma

Department of Computer Science and Engineering
Shri Jagdishprasad Jhabarmal Tibrewala University
Jhunjhunu, Rajasthan, India
dr.sharmayogeshkumar@gmail.com

Citation:

Please cite the final published version of this work:

```
@INPROCEEDINGS{11368510,
  author={Kalamkar, Pratik N. and Sharma, Yogesh Kumar},
  booktitle={2025 IEEE 6th Global Conference for Advancement in Technology
     (GCAT)},
  title={Hierarchical Ordinal Framework for Automated Movie Censorship
     Using Full-Length Scripts},
  year={2025},
  volume={},
  number={},
  pages={1-7},
  keywords={Sensitivity;Reviews;Entertainment industry;Benchmark testing;
     Predictive models;Motion pictures;Transformers;Censorship;
     Certification;Context modeling;component;formatting;style;styling;
     insert},
  doi={10.1109/GCAT66372.2025.11368510}}
```


Abstract

The certification of movies into age-appropriate categories is traditionally performed by human censorship boards, but this process is often subjective, inconsistent, and impractical at the scale demanded by digital streaming platforms. Existing automated methods rely primarily on subtitles or user-generated reviews, which provide only partial representations of film content, and they typically treat certification as a flat classification problem, ignoring the ordered structure of rating categories

(G, PG, PG-13, R, NC-17). In this study, we propose a novel framework for automatic movie certification using full-length scripts, capturing both dialogue and narrative descriptions that influence censorship judgments. To overcome input length constraints, we adopt a hierarchical chunking strategy that preserves full-script coverage while enabling model processing. Furthermore, we integrate ordinal regression layers into sequential (HLSTM) and transformer-based (BERT, Longformer) models to enforce order-consistent predictions, and benchmark them against a TF-IDF baseline under balanced and imbalanced settings. Experimental results on a dataset of 1,142 English scripts reveal that while TF-IDF baselines remain surprisingly robust in low-resource conditions (64.5% accuracy, QWK > 0.72), deep contextual models demonstrate sensitivity to class balance and hold promise for larger-scale deployments. This work makes the first comprehensive benchmark of censorship classification on full scripts, highlighting when simple models outperform sophisticated architectures, and providing both methodological insights and policy relevance for scalable, automated censorship frameworks in the film industry.

Keywords

Natural Language Processing (NLP), Text Classification, Hierarchical Learning, Text Classification, Movie Script Analysis

1 Introduction

The certification of movies into age-appropriate categories (e.g., G, PG, PG-13, R, NC-17) is an essential regulatory process intended to guide audiences and protect vulnerable groups. Traditionally, this task has been carried out manually by censorship boards, relying on subjective interpretation of violence, explicit content, and sensitive themes. With the exponential growth of digital media and streaming platforms, the demand for rapid and consistent certification has increased dramatically, exposing the limitations of manual approaches in terms of scalability, consistency, and objectivity. This has motivated research into automated methods for predicting movie ratings, which could support or augment human decision-making [1]

Despite progress in text classification, most prior research related to movie content analysis has focused on subtitles or user-generated reviews, which provide only partial representations of a film's narrative. Subtitles omit crucial descriptive elements such as violent actions, sexual references, or implied themes, all of which play a significant role in censorship decisions. Moreover, existing approaches typically frame the problem as a flat classification task, ignoring the ordinal structure of certification categories. Misclassifying a "G" movie as "PG" is far less severe than misclassifying it as "R," yet conventional models treat both errors equally. These limitations underscore the need for a more comprehensive and contextually sensitive approach [2, 3].

In this study, we address these gaps by proposing a novel framework for automatic movie certification using full-length scripts. Unlike prior subtitle-based studies, our approach captures both dialogue and narrative descriptions, thereby preserving the contextual richness that influences human censorship judgments. We further introduce ordinal regression layers into traditional, recurrent, and transformer-based models, ensuring that predictions align with the ordered nature of censorship categories. Specifically, we compare four complementary architectures: (i) TF-IDF with classical classifiers, (ii) Hierarchical LSTM (HLSTM) with ordinal regression, (iii) BERT with ordinal regression, and (iv) Longformer with ordinal regression for long-sequence modeling.

The key contributions of this research are, First use of full-length scripts for censorship classification, moving beyond subtitles or metadata. Integration of ordinal regression into both sequential and transformer-based models for more interpretable and order-consistent predictions. Comprehensive benchmarking of classical, recurrent, and transformer models under balanced and imbalanced data conditions, revealing critical insights into when simple models outperform sophisticated architectures. Policy and industry relevance, demonstrating the feasibility of scalable, automated censorship classification for streaming platforms and certification authorities.

While prior studies have made significant contributions to text classification, several limitations persist: (i) reliance on short text units (subtitles, paragraphs, or legal snippets) rather than full-length documents; (ii) lack of explicit modeling of ordinal labels, particularly relevant for tasks like censorship ratings where categories (G, PG, PG-13, R, NC-17) follow a natural order; and (iii) insufficient benchmarking across both traditional and deep learning models under the same experimental setup. This study addresses these gaps by proposing an end-to-end framework for automatic movie certification using full-length scripts rather than subtitles, thereby incorporating both dialogue and narrative cues. To avoid truncation imposed by model input limits, we introduce a hierarchical modeling strategy: each script is divided into smaller chunks that inherit the same certification label as the parent script, and predictions from these chunks are then aggregated at the document level. This design ensures that the system leverages the entirety of the script while maintaining fidelity to censorship labels at the script level. Moreover, by integrating ordinal regression layers with models such as HLSTM, BERT, and Longformer, we align predictions with the inherent order of censorship categories. Finally, by systematically comparing TF-IDF, HLSTM, BERT, and Longformer under balanced and unbalanced settings, this work provides one of the first comprehensive benchmarks for censorship classification, highlighting both the strengths of classical methods in low-resource conditions and the potential of deep models when scaled.

This paper is organized as follows: Section 2 reviews related work on text classification and movie rating prediction. Section 3 details the methodology and proposed framework, while Section 4 describes the experimental setup. Section 5 presents results and discussions, followed by Section 6, which concludes the study and outlines directions for future work.

2 Related Work / Literature Review

Text classification has been one of the most widely studied problems in natural language processing (NLP), and numerous approaches ranging from classical machine learning to advanced deep learning models have been proposed. Our earlier work explored sentiment-based approaches for script classification, combining lexicon-based models such as VADER and TextBlob to approximate censorship ratings [4]. While this demonstrated the feasibility of lightweight methods, performance remained limited due to the models' inability to capture long-range dependencies and narrative cues. Subsequently, we conducted a systematic survey of movie automation research [5], which identified gaps such as reliance on short texts (subtitles, metadata) and the underuse of full scripts. The present study builds upon these insights, introducing a hierarchical ordinal framework over full-length scripts and benchmarking both traditional and transformer-based architectures.

Starting early work relied heavily on Bag-of-Words (BoW) and TF-IDF features with classifiers such as Support Vector Machines (SVMs) and Multi-Layer Perceptrons (MLPs). For instance, Are We Really Making Much Progress? [6] benchmarked BoW against neural and transformer models, showing that while BoW methods remain competitive in certain cases, they often fail to capture contextual and semantic nuances. Similarly, studies such as Text Classification Neural [1] compared MLPs, RNNs, and SVMs, reinforcing the trend that shallow models, while efficient, are limited in modeling long-range dependencies. With the advent of deep learning, sequence-based models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), including LSTM and GRU variants, became dominant. Several comparative studies—such as [2, 7–9] evaluated CNNs, RNNs, BiLSTMs, and GRUs across different text classification domains. These works demonstrated that CNNs are effective in extracting local n-gram features, while RNNs and LSTMs excel in modeling sequential dependencies. However, they also revealed limitations in scalability and handling very long documents, which is particularly relevant for movie scripts. The introduction of transformer-based models such as BERT, RoBERTa, DistilBERT, ALBERT, and XLNet [10–14] revolutionized text classification. Fine-tuning pretrained transformers significantly improved performance across multiple benchmarks, especially for tasks requiring nuanced semantic understanding. Comparative evaluations such as A Fine-Tuned BERT-Based Transfer Learning Approach for Text Classification [14] and Enhancing Indian Legal Text Classification [15] demonstrated the superiority of transformer models over CNNs and RNNs, particularly in domain-specific applications such as legal and technical document classification. However, these models typically struggle with long documents due to input length restrictions (e.g., 512 tokens in BERT). To address this limitation, extended transformer architectures such as Longformer, BigBird, and LUKE were introduced and tested in works like [3], [16] and [17].These models improved long-text handling but still required careful preprocessing (e.g., chunking, hierarchical structures) and large-scale training resources. Beyond flat classification, some researchers explored hierarchical and attention-based models. For example, Hierarchical Attention and Transformers for Automatic Movie Rating [18] proposed hierarchical LSTMs and hierarchical transformers for rating prediction, showing that preserving document structure significantly enhances performance. Similarly, Deep Learning for Technical Document Classification [19] and Hierarchical Neural Network Approaches [16] employed Hierarchical Attention Networks (HANs) and Longformer-based architectures to better capture context across long documents. While effective, these methods require substantial computational resources and may overfit in small datasets. Work has also been done specifically in movie-related text analysis. For example, Prediction of Motion Picture Association Ratings [20] combined lexical features (TF-IDF), emotional features, and deep learning architectures such as LSTMs for rating prediction. Likewise, Anticipate Movie Theme from Subtitle [21] employed CNNs, RNNs, LSTMs, and GRUs on subtitle data for thematic classification. While these studies highlight the feasibility of automated classification in the film domain, they largely depend on subtitles rather than full scripts, thus ignoring descriptive and narrative elements that are central to human censorship decisions. Similarly, Movie Genre Classification Using Neural Networks and Transformers [22] focused on genre categorization but did not address the regulatory perspective of censorship. Survey and comparative studies [7, 12, 13, 17, 23] have consistently emphasized that while transformer-based models outperform traditional approaches, their performance degrades when input texts are significantly longer than the token limit. This is especially critical in domains like legal and cinematic texts, where document lengths can span thousands of tokens.

In addition, several recent studies provide further insights into text classification across diverse domains. Deep learning architectures such as FFNNs, CNNs, RNNs, and LSTMs have demonstrated strong performance in general text classification tasks [8]. Multilingual categorization studies employing models like mBERT and XLM-RoBERTa achieved promising results across multiple languages, highlighting their applicability to multilingual censorship contexts [24]. Domain-specific applications, such as multi-label legal document classification using BIGRU-LWAN, AttentionXML, and X-Transformer, illustrate the effectiveness of specialized architectures in complex regulatory tasks [25].

3 Methodology / Proposed Approach

This study presents a novel framework for automatic movie certification by analyzing full-length English movie scripts and classifying them into censorship ratings (G, PG, PG-13, R, NC-17). Unlike prior approaches that rely mainly on subtitles or metadata, we utilize complete scripts, which contain both narrative descriptions and dialogues, thereby preserving crucial contextual cues often considered in human certification processes. The central novelty of our approach lies in the integration of ordinal regression with deep learning architectures to capture the natural order of censorship ratings, coupled with a systematic comparison of classical and modern text classification techniques.

The preprocessing pipeline was designed to convert raw script files into model-ready input while retaining important narrative elements. Scripts were parsed, and relevant metadata such as titles and certification labels were extracted. The text was normalized by converting to lowercase, removing extraneous symbols, and applying lemmatization to reduce morphological variations. Unlike standard preprocessing approaches that flatten text, we preserved scene and dialogue markers, which ensured that sequential and hierarchical models could exploit structural dependencies. Given that movie scripts are significantly longer than the maximum input length of standard transformer models, we adopted a chunking strategy. For BERT, the scripts were divided into overlapping windows of 512 tokens, with predictions aggregated through weighted averaging. For Longformer, which supports extended input lengths, the chunking was minimal, thereby preserving narrative coherence across scenes. Furthermore, class imbalance was addressed by applying class weights during training and ensuring stratified splits of training, validation, and test sets.

We implemented four models to capture different perspectives of script representation and classification. As a baseline, we used TF-IDF features combined with classical classifiers such as Logistic Regression and Support Vector Machines. This provided a sparse representation of text and established a reference point for performance evaluation. We also implemented a Hierarchical LSTM (HLSTM), where sentence-level embeddings were first generated and then aggregated at the script level, capturing both local and global dependencies within the narrative. To integrate the ordered nature of censorship labels, we employed an ordinal regression output layer, which penalized large deviations in predictions (e.g., predicting G as R) more heavily than minor misclassifications (e.g., G as PG).

In addition, we fine-tuned BERT on the movie script dataset. Due to input length restrictions, we trained BERT on script segments, and final predictions were derived by aggregating segment-level outputs. As with HLSTM, we replaced the standard softmax classifier with an ordinal regression layer, ensuring consistency with the hierarchical structure of censorship ratings. Alongside TF-IDF, HLSTM, and BERT, we also evaluated Longformer, a transformer model designed for long documents through sparse attention mechanisms. By processing up to 4,096 tokens per chunk, Longformer was particularly suited for this task, as it preserved long-range narrative dependencies that are otherwise truncated in standard transformers. An ordinal regression output layer was similarly applied to Longformer to align predictions with the ordered nature of the certification scheme.

Theoretically, our use of ordinal regression was inspired by cumulative link models, where the probability of a script belonging to a particular class y ¡= k is modeled as in equation 1.

$$P(y \leq k \mid x) = \sigma(\theta_k - f(x)) \tag{1}$$

with f(x) representing the model output, θ_k denoting learned thresholds, and σ being the logistic sigmoid function. The predicted class is obtained as the smallest k for which the cumulative probability exceeds 0.5. This formulation enforces the ordinal nature of labels, ensuring that predictions are monotonic and interpretable.

All models were trained under a unified strategy to ensure fair and consistent evaluation across approaches. Optimization was carried out using Adam or AdamW optimizers with model-specific learning rates tailored to each architecture. To address class imbalance, class-weighted loss functions were employed in conjunction with weighted random sampling, thereby improving robustness on underrepresented classes. Regularization techniques included early stopping with a patience of three epochs based on validation Quadratic Weighted Kappa (QWK), dropout layers, and gradient clipping to prevent overfitting and stabilize training. For transformer-based models, fine-tuning was further enhanced through discriminative learning rates and selective layer freezing, enabling efficient adaptation while retaining the benefits of pre-trained representations.

In summary, the design choices were motivated by both technical and practical considerations. Full scripts were chosen over subtitles because they capture content elements such as implied violence and narrative description, which strongly influence censorship decisions but are often omitted from subtitles. Ordinal regression was adopted to align the model outputs with the inherent ordered nature of movie ratings. Longformer was selected to address the challenge of modeling long texts without excessive truncation, while multiple models were benchmarked to highlight trade-offs between interpretability, efficiency, and performance.

4 Experimental Setup

The experiments were conducted on a curated collection of English-language movie scripts sourced from IMDB and Scripts.com, once we had movie scripts, missing and existing MPAA ratings were filled and verified with help python script running on OMDB. The full unbalanced dataset comprised approximately 1142 scripts spanning five censorship categories: G, PG, PG-13, R, and NC-17. The balanced dataset consists of 250 scipts, 50 for each censorship category. In case of unbalanced dataset the distribution was inherently imbalanced, with PG and PG-13 dominating, while NC-17 and G were underrepresented. Preprocessing steps, as outlined earlier, were applied uniformly across the dataset to ensure consistency. Stratified splits were used to divide the data into training, validation, and test sets, thereby preserving class proportions and mitigating bias.

To ensure consistency and comparability across different modeling approaches, all four prototypes—TF-IDF, HLSTM, BERT, and Longformer—were developed under a unified experimental pipeline, Fig. 1. The methodology comprises the following stages:

4.1 Data Acquisition and Preprocessing

Raw movie scripts were collected and subjected to standard text preprocessing, including removal of special characters, normalization, and tokenization. The dataset was then partitioned into training, validation, and test splits to enable robust evaluation. Rating labels were encoded into numeric form to support supervised learning.

4.2 Handling Class Imbalance

In case of unbalanced dataset, given the natural skew in movie certification categories, imbalance mitigation strategies were applied uniformly across all experiments. These included class-weighted loss functions and weighted random sampling during training to prevent majority-class bias.

4.3 Feature Representation / Input Encoding

Each prototype employed a different representation strategy, but all shared the common principle of transforming raw text into machine-processable features:

- **TF-IDF:** Statistical n-gram features.
- **HLSTM:** Word embeddings capturing sequential context.
- **BERT:** Contextualized embeddings from a pre-trained transformer.
- **Longformer:** Chunked long-sequence embeddings with transformer attention.

4.4 Model Design

The models were optimized for movie certification classification. Deep learning and transformer-based approaches (HLSTM, BERT, Longformer) incorporated ordinal regression layers, aligning with the ordered nature of certification ratings, whereas the TF-IDF model employed categorical classification as a baseline.

4.5 Training Strategy

A consistent training strategy was maintained across all prototypes. Table 1, shows key parameters.

- **Optimizers:** Adam or AdamW with model-specific learning rates.
- **Regularization:** Early stopping (patience = 3) based on validation Quadratic Weighted Kappa (QWK), dropout layers, and gradient clipping.
- **Fine-tuning:** For transformer-based models, discriminative learning rates and selective layer freezing were employed to balance generalization and task adaptation.

4.6 Evaluation Metrics

To ensure holistic performance assessment, multiple evaluation measures were adopted, including: Accuracy, Macro-Precision, Macro-Recall, Macro F1-score, Mean Absolute Error (MAE), Quadratic Weighted Kappa (QWK), Spearman's correlation coefficient (ρ). This multi-metric approach facilitated fair comparison across models with varying architectures.

4.7 Comparative Analysis

Results from all four prototypes were benchmarked within the same experimental framework, enabling systematic comparison of classical feature-based, sequential deep learning, and transformer-based methods.

Table 1: Key Model Hyperparameters

Model	**Key Hyperparameters**
TF-IDF+MLP	`ngram_range=(1,2)`, `max_features=80,000`, MLP layers: `[100, 50]`, Dropout=`0.5`
HLSTM	`vocab_size=10,000`, `embedding_dim=128`, `chunk_len=200`, `max_chunks=10`
Hierarchical BERT	`MAX_CHUNK_SIZE=512`, `OVERLAP=50`, 6 frozen BERT layers, LR (BERT): `5e-5`, LR (Head): `1e-4`
Hierarchical Longformer	`chunk_size=4096`, Unfrozen layers: last 2, Script Aggregator: 2-layer Transformer, LR (Longformer): `1e-5`, LR (Head): `5e-4`

All experiments were executed in a controlled computational environment. Training was carried out on NVIDIA Tesla V100 and T4 GPUs with 16–32 GB of memory. The software environment consisted of Python 3.10, with PyTorch 2.1 powering BERT and Longformer models, TensorFlow/Keras implementing the HLSTM architecture, and scikit-learn supporting TF-IDF feature extraction and baseline classification models. Tokenization and model loading for transformers were handled using the HuggingFace Transformers library.

Model training followed consistent hyperparameter settings where possible. For deep learning models, the AdamW optimizer was used, with a learning rate of 2×10^{-5} for transformer-based models and 1×10^{-3} for HLSTM and TF-IDF baselines. Batch sizes were set to 16 for transformer models and 64 for lighter models, while training epochs ranged from 10 to 15 with early stopping based on validation loss (patience = 3). For the baseline models, cross-entropy loss was employed, whereas the deep learning models utilized a custom ordinal regression loss function to reflect the ordered nature of the output classes.

Performance was evaluated using a diverse set of metrics to capture both classification accuracy and ordinal consistency. Overall correctness was measured using accuracy, while macro precision, recall, and F1-scores were employed to ensure fairness across imbalanced classes. Since censorship ratings follow a natural order, Mean Absolute Error (MAE) was computed to measure the severity of misclassification in ordinal space. To further assess alignment between predictions and ground truth, we used Quadratic Weighted Kappa (QWK), which penalizes larger disagreements more heavily, and Spearman's rank correlation coefficient, which measures the monotonic relationship between predicted and actual ratings.

The combination of these experimental strategies enabled a comprehensive evaluation of our proposed framework. By systematically comparing TF-IDF, HLSTM, BERT, and Longformer, we highlight not only the performance gains of deep contextual models but also the necessity of treating movie ratings as an ordinal prediction task rather than a flat classification problem.

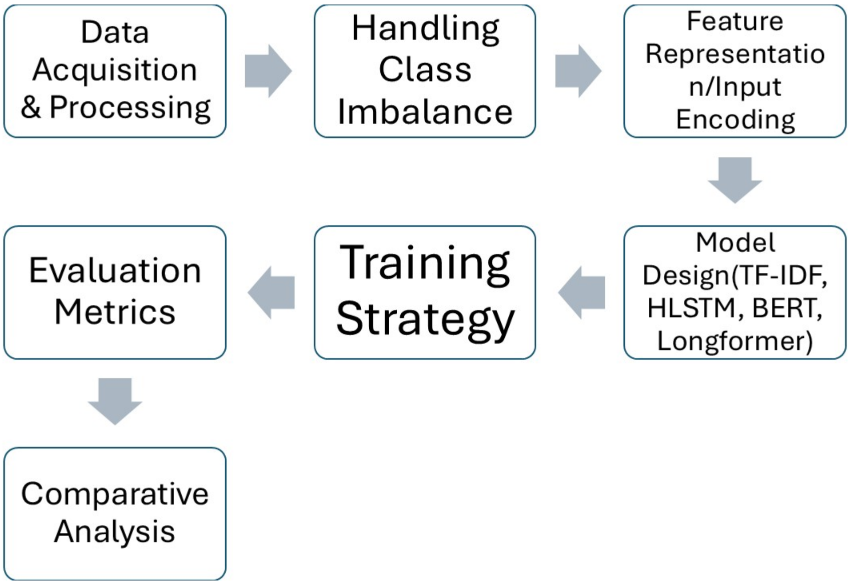

Figure 1: Experimental Workflow

5 Results and Discussion

The performance of the proposed models was evaluated on both unbalanced and balanced datasets. Figure 2 summarize the comparative results across TF-IDF, HLSTM, BERT, and Longformer, while Fig. 3 and Fig. 4 (confusion matrices) illustrate misclassification patterns for TF-IDF as a representative case.

Across both settings, the TF-IDF baseline consistently outperformed the more complex deep learning models. In the unbalanced dataset, TF-IDF achieved an accuracy of 64.5% with a QWK of 0.7273, while in the balanced dataset it maintained stable performance with 60.5% accuracy and 0.7714 QWK. This suggests that simple sparse representations are surprisingly robust in low-resource settings, where transformers often underperform due to insufficient training data.

The confusion matrices provide insight into the types of errors made by the models. The unbalanced matrix shows a clear dominance of R and PG-13 predictions, reflecting dataset skew. However, in the balanced dataset, classification is more evenly distributed, with G, NC-17, and PG also showing good recall. For example, TF-IDF correctly classified 15/25 G-rated scripts and 13/25 NC-17 scripts, showing that even simple models capture strong discriminative features when class proportions are normalized.

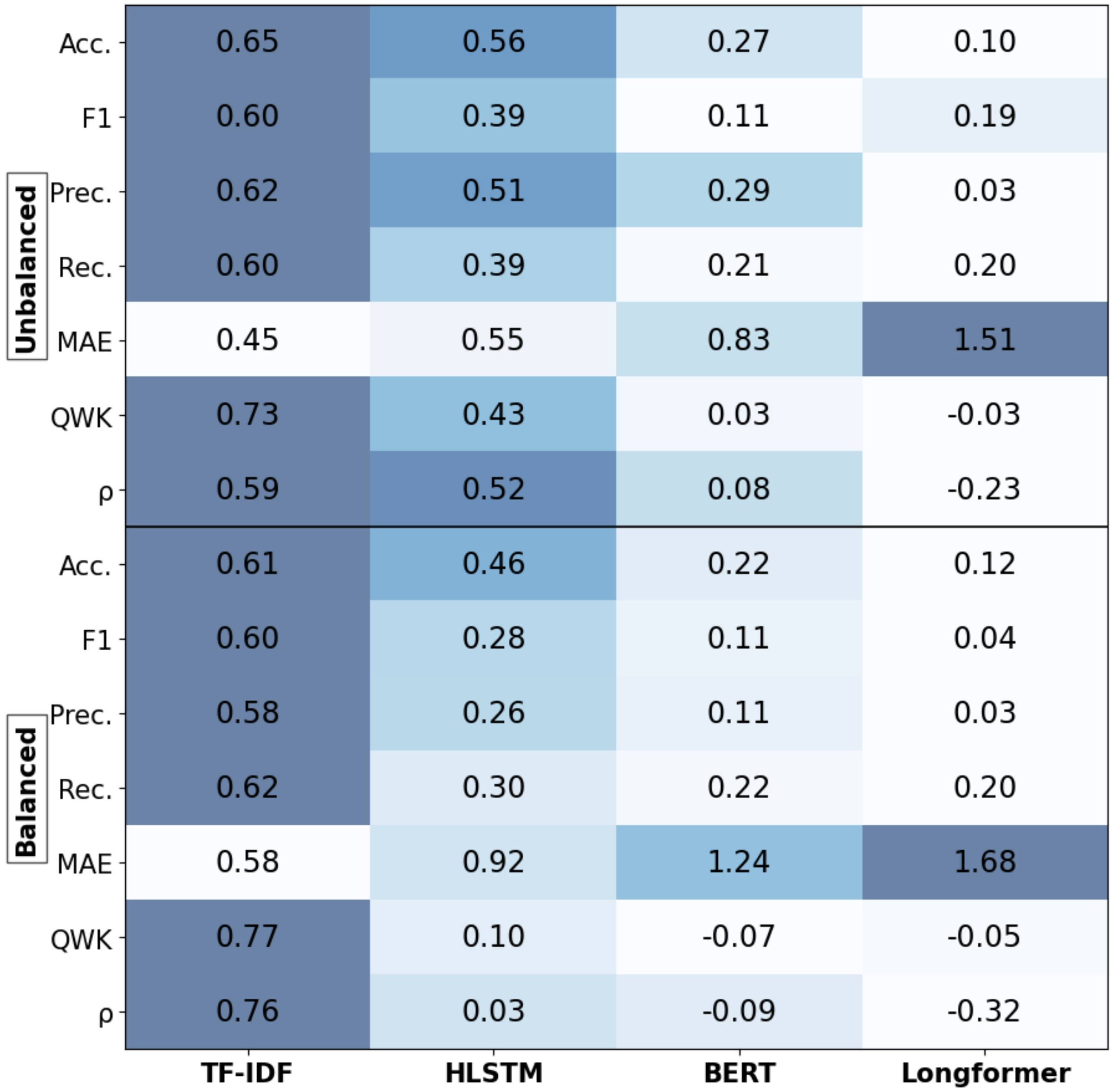

Figure 2: Overall performance metrics comparison across models (TF-IDF, HLSTM, BERT, Longformer).

These results, Fig, 4, highlight two key insights. First, TF-IDF demonstrates stability across both imbalanced and balanced conditions, making it a reliable choice in low-resource domains. Second, the confusion matrices reveal that while transformers struggle with small datasets, they still occasionally capture subtle boundaries, suggesting potential for improvement with larger corpora.

Despite the comparatively lower performance of HLSTM, BERT, and Longformer models, the results highlight important considerations and potential directions. First, it must be noted that the system ingests entire movie scripts as raw input, and with only the overall rating serving as the supervisory signal. This inherently coarse granularity imposes a challenging learning setup, where even powerful transformer architectures may struggle to capture fine-grained cues. Nonetheless, the transformers demonstrated noticeable sensitivity to class balancing, with performance shifts across unbalanced and balanced datasets indicating their adaptability to improved data distributions. Furthermore, confusion matrix patterns suggest that these models were able to capture strong signals for dominant categories such as R and PG-13, underscoring their potential to learn discriminative features when data is more representative. While sparse representations like TF-IDF remain surprisingly robust in low-resource conditions, the richer contextual modeling capacity of transformers positions them as strong candidates for future work, particularly when larger, more diverse corpora or ensemble strategies are employed.

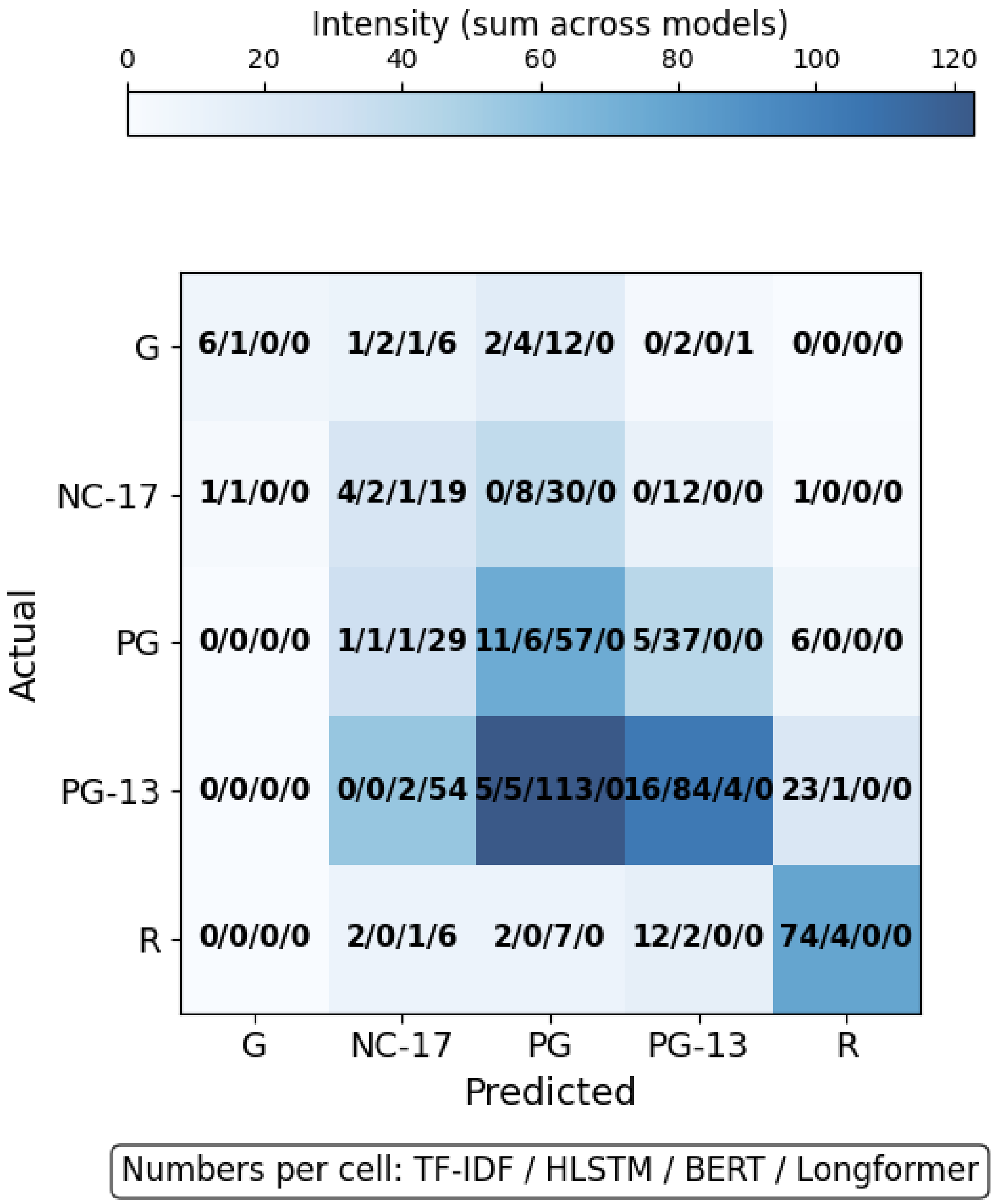

Figure 3: Confusion matrix of models on the unbalanced dataset (with class weights).

6 Conclusion and Future Work

Building on our earlier findings with lexicon-based sentiment analysis [4] and the trends highlighted in our comprehensive survey [5], this work demonstrates the advantages of full-script modeling and ordinal regression with hierarchical modelling. We proposed and evaluated a framework for automatic movie certification using full-length scripts, introducing ordinal regression layers into both recurrent and transformer-based models. The work leverages complete scripts—rather than subtitles or metadata—for censorship classification, thereby preserving narrative and descriptive cues that influence certification decisions.

Experimental results across multiple architectures demonstrate that TF-IDF baselines outperform deep learning models under current dataset constraints, achieving the highest accuracy (64.5% unbalanced, 60.5% balanced) and strongest ordinal alignment (QWK > 0.72).

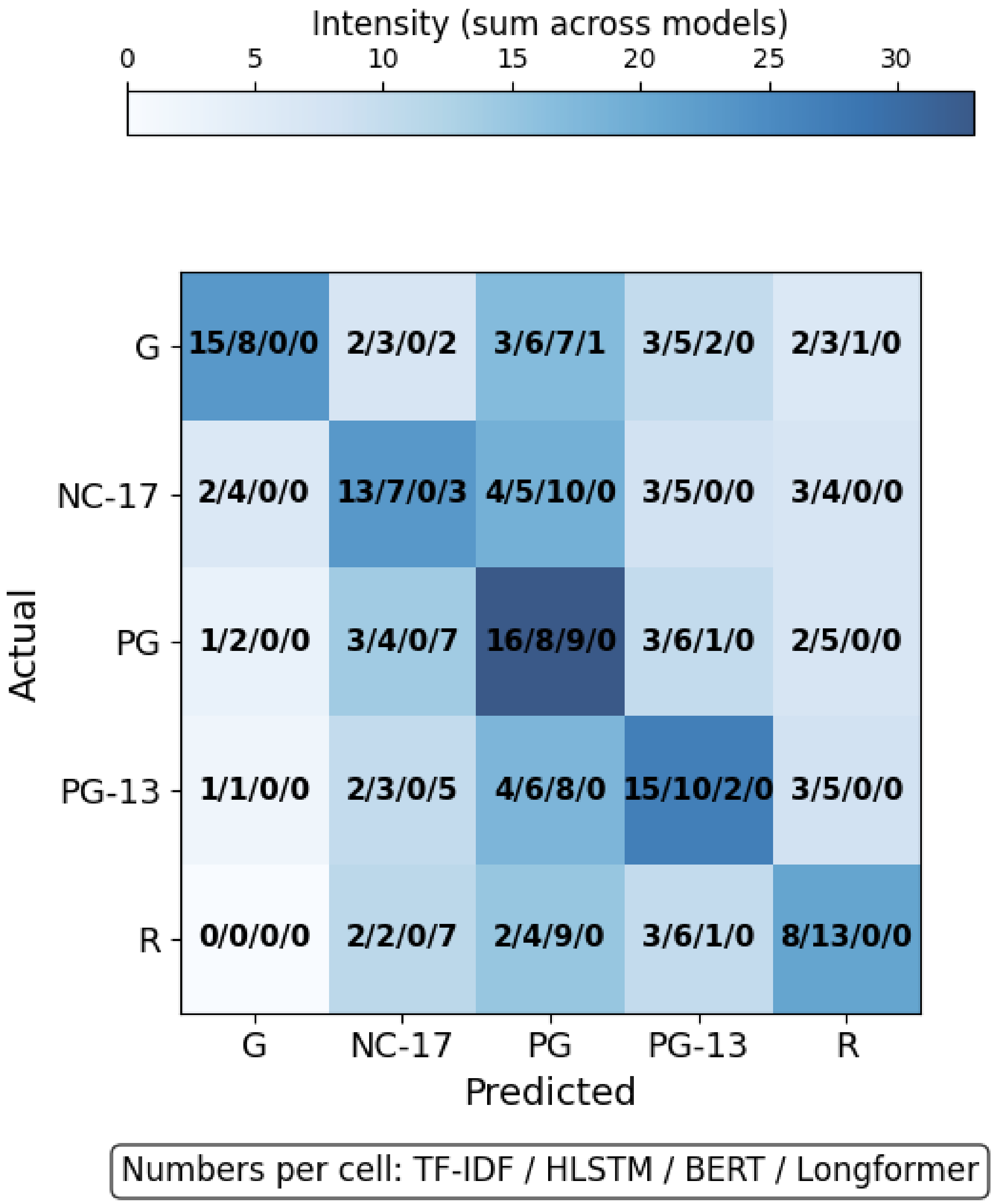

Figure 4: Confusion matrix of models on the balanced dataset.

A key factor contributing to the relatively poor performance of transformer models in this study lies in the nature of the input representation. The system processes entire scripts in raw form, assigning only a single global certification label per document. This setup introduces a weak supervision problem, as the majority of the script content (neutral dialogue, filler narrative, or descriptive text) does not directly align with the rating decision, thereby diluting the signal available to deep contextual models. Transformers such as BERT and Longformer, while designed for capturing fine-grained sequential dependencies, are particularly sensitive to input length and label granularity, and often require more targeted supervision to fully exploit their representational capacity. In contrast, TF-IDF benefits from its sparse frequency-based representation, where even infrequent but rating-indicative terms exert strong discriminative influence across the entire document. Thus, the relative advantage of TF-IDF in this setting does not indicate a structural weakness of transformers per se, but rather reflects the inherent challenge of full-script classification under coarse labeling.

Interestingly, despite applying chunking and fine-tuning strategies, transformer models (BERT, Longformer) yielded very low or even negative Spearman correlations ($\leq$0.08 unbalanced, $\leq$-0.32 balanced), indicating a lack of ordinal alignment between predictions and ground truth. This suggests that, under current dataset size and domain-specific constraints, transformers may struggle to model the global narrative flow necessary for certification prediction, in contrast to simpler TF-IDF representations which preserved ranking consistency.

In conclusion, this work contributes a novel perspective on movie certification by reframing it as an ordinal text classification task, provides a benchmark comparison across classical and deep learning models, and highlights practical insights into when simple models can outperform sophisticated architectures. These results offer both methodological and policy relevance, laying the groundwork for scalable, automated censorship classification in the future.

Additionally, future research could explore domain-specific pretraining, semi-supervised learning, and cross-lingual extensions to Hindi and Marathi scripts, enabling multilingual certification frameworks. Incorporation of multimodal cues (e.g., video or audio features) could also create more holistic censorship prediction systems.

References

[1] C. Petridis, "Text Classification: Neural Networks VS Machine Learning Models VS Pre-trained Models," Dec. 2024, arXiv:2412.21022 [cs]. [Online]. Available: http://arxiv.org/abs/2412.21022

[2] H. Lu, L. Ehwerhemuepha, and C. Rakovski, "A comparative study on deep learning models for text classification of unstructured medical notes with various levels of class imbalance," *BMC Medical Research Methodology*, vol. 22, no. 1, p. 181, Dec. 2022. [Online]. Available: https://bmcmedresmethodol.biomedcentral.com/articles/10.1186/s12874-022-01665-y

[3] L. M. Pham and H. Cao The, "LNLF-BERT: Transformer for Long Document Classification With Multiple Attention Levels," *IEEE Access*, vol. 12, pp. 165 348–165 358, 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10744540/

[4] P. N. Kalamkar and Y. K. Sharma, "Blend of VADER and TextBlob for Movie Script Classification," in *2024 4th International Conference on Artificial Intelligence, Robotics, and Communication (ICAIRC)*. Xiamen, China: IEEE, Dec. 2024, pp. 875–881. [Online]. Available: https://ieeexplore.ieee.org/document/10899931/

[5] P. N. Kalamkar and Y. Kumar Sharma, "A review of research trends in movie automation," in *2025 International Conference on Computer Technology Applications (ICCTA)*. IEEE, May 2025, pp. 154–163. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/11166207

[6] L. Galke, A. Scherp, A. Diera, F. Karl, B. X. Lin, B. Khera, T. Meuser, and T. Singhal, "Are We Really Making Much Progress in Text Classification? A Comparative Review," Jan. 2025, arXiv:2204.03954 [cs]. [Online]. Available: http://arxiv.org/abs/2204.03954

[7] V. Dogra, S. Verma, Kavita, P. Chatterjee, J. Shafi, J. Choi, and M. F. Ijaz, "A Complete Process of Text Classification System Using State-of-the-Art NLP Models," *Computational Intelligence and Neuroscience*, vol. 2022, pp. 1–26, Jun. 2022. [Online]. Available: https://www.hindawi.com/journals/cin/2022/1883698/

[8] P. Lavanya and E. Sasikala, "Deep Learning Techniques on Text Classification Using Natural Language Processing (NLP) In Social Healthcare Network: A Comprehensive Survey," in *2021 3rd International Conference on Signal Processing and Communication (ICPSC)*. Coimbatore, India: IEEE, May 2021, pp. 603–609. [Online]. Available: https://ieeexplore.ieee.org/document/9451752/

[9] B. Rogers, N. Noman, S. Chalup, and P. Moscato, "A comparative analysis of deep neural network architectures for sentence classification using genetic algorithm," *Evolutionary Intelligence*, vol. 17, no. 3, pp. 1933–1952, Jun. 2024. [Online]. Available: https://link.springer.com/10.1007/s12065-023-00874-8

[10] A. Areshey and H. Mathkour, "Exploring transformer models for sentiment classification: A comparison of BERT, RoBERTA, AlBERT, DistilBERT and XLNet," *Expert Systems*, vol. 41, no. 11, p. e13701, Nov. 2024. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1111/exsy.13701

[11] A. Esmaeilzadeh and K. Taghva, "Text Classification Using Neural Network Language Model (NNLM) and BERT: An Empirical Comparison," in *Intelligent Systems and Applications*, K. Arai, Ed. Cham: Springer International Publishing, 2022, vol. 296, pp. 175–189, series Title: Lecture Notes in Networks and Systems. [Online]. Available: https://link.springer.com/10.1007/978-3-030-82199-9-12

[12] A. Gasparetto, M. Marcuzzo, A. Zangari, and A. Albarelli, "A Survey on Text Classification Algorithms: From Text to Predictions," *Information*, vol. 13, no. 2, p. 83, Feb. 2022. [Online]. Available: https://www.mdpi.com/2078-2489/13/2/83

[13] Q. Li, H. Peng, J. Li, C. Xia, R. Yang, L. Sun, P. S. Yu, and L. He, "A Survey on Text Classification: From Traditional to Deep Learning," *ACM Transactions on Intelligent Systems and Technology*, vol. 13, no. 2, pp. 1–41, Apr. 2022. [Online]. Available: https://dl.acm.org/doi/10.1145/3495162

[14] R. Qasim, W. H. Bangyal, M. A. Alqarni, and A. Ali Almazroi, "A Fine-Tuned BERT-Based Transfer Learning Approach for Text Classification," *Journal of Healthcare Engineering*, vol. 2022, pp. 1–17, Jan. 2022. [Online]. Available: https://www.hindawi.com/journals/jhe/2022/3498123/

[15] P. Prabhakar and P. B. Pati, "Enhancing Indian legal judgment classification with embeddings, feature selection, and ensemble strategies," *Artificial Intelligence and Law*, Mar. 2025. [Online]. Available: https://link.springer.com/10.1007/s10506-025-09438-w

[16] S. I. Khandve, V. K. Wagh, A. D. Wani, I. M. Joshi, and R. B. Joshi, "Hierarchical Neural Network Approaches for Long Document Classification," in *2022 14th International Conference on Machine Learning and Computing (ICMLC)*. Guangzhou China: ACM, Feb. 2022, pp. 115–119. [Online]. Available: https://dl.acm.org/doi/10.1145/3529836.3529935

[17] V. Wagh, S. Khandve, I. Joshi, A. Wani, G. Kale, and R. Joshi, "Comparative Study of Long Document Classification," in *TENCON 2021 - 2021 IEEE Region 10 Conference (TENCON)*. Auckland, New Zealand: IEEE, Dec. 2021, pp. 732–737. [Online]. Available: https://ieeexplore.ieee.org/document/9707465/

[18] L. F. Pardo-Sixtos, A. P. López-Monroy, M. Shafaei, and T. Solorio, "Hierarchical attention and transformers for automatic movie rating," *Expert Systems with Applications*, vol. 209, p. 118164, Dec. 2022. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0957417422013240

[19] S. Jiang, J. Hu, C. L. Magee, and J. Luo, "Deep Learning for Technical Document Classification," *IEEE Transactions on Engineering Management*, vol. 71, pp. 1163–1179, 2024. [Online]. Available: https://ieeexplore.ieee.org/document/9729968/

[20] G. D. Indiran, "Prediction of motion picture association ratings using emotion analysis and text classification approaches," M.Sc. thesis, Tilburg University, Tilburg, The Netherlands, Sep 2021.

[21] Y. I. Alzoubi, A. E. Topcu, E. Elbasi, M. Buyukyilmaz, and A. O. Cibikdiken, "Anticipate Movie Theme from Subtitle: A Deep Learning Approach," in *2024 47th International Conference on Telecommunications and Signal Processing (TSP)*. Prague, Czech Republic: IEEE, Jul. 2024, pp. 205–210. [Online]. Available: https://ieeexplore.ieee.org/document/10605925/

[22] M. R. Cuomo, "Movie genre classification using script texts," M.Sc. thesis, Florida Atlantic University, Boca Raton, Florida, USA, May 2025, electronic Theses and Dissertations, #80. [Online]. Available: https://digitalcommons.fau.edu/\discretionary{}{}{}etd_general/80/

[23] H. Chen, L. Wu, J. Chen, W. Lu, and J. Ding, "A comparative study of automated legal text classification using random forests and deep learning," *Information Processing & Management*, vol. 59, no. 2, p. 102798, Mar. 2022. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0306457321002764

[24] G. Manias, A. Mavrogiorgou, A. Kiourtis, C. Symvoulidis, and D. Kyriazis, "Multilingual text categorization and sentiment analysis: a comparative analysis of the utilization of multilingual approaches for classifying twitter data," *Neural Computing and Applications*, vol. 35, no. 29, pp. 21 415–21 431, Oct. 2023. [Online]. Available: https://link.springer.com/10.1007/s00521-023-08629-3

[25] D. Song, A. Vold, K. Madan, and F. Schilder, "Multi-label legal document classification: A deep learning-based approach with label-attention and domain-specific pre-training," *Information Systems*, vol. 106, p. 101718, May 2022. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0306437921000016